

Md Mobarak Karim

Biomedical Optics Researcher | Postdoctoral Candidate

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Academic Profile

Biomedical optics researcher with a Ph.D. in Biomedical Engineering and more than six years of experience developing multimodal optical imaging systems for developmental biology. Research expertise includes optical coherence tomography (OCT), one- and two-photon light-sheet fluorescence microscopy (LSFM), two-photon microscopy, quantitative tissue optics, optical system integration, and computational image analysis. Led the design, validation, and biological application of co-registered OCT-LSFM platforms for mouse and zebrafish embryo imaging, resulting in first-author journal publications, SPIE proceedings, and competitive research and travel awards.

Research Expertise

Biomedical Imaging	Swept-source OCT, spectral-domain OCT, OCT angiography, optical coherence elastography, one-photon LSFM, two-photon LSFM, multiphoton microscopy, confocal and fluorescence microscopy
Optical System Development	Interferometer design, optical alignment, Gaussian beam propagation, beam expanders, 4f relays, galvo scanning, scan/tube lenses, fiber optics, polarization control, laser-camera synchronization, incubator integration
Quantitative Imaging	Attenuation-coefficient mapping, sensitivity roll-off correction, confocal correction, image quality metrics, morphometry, segmentation, multimodal co-registration, 3D/4D reconstruction, statistical analysis
Programming and Software	MATLAB, Python, C/C++, LabVIEW, Zemax OpticStudio, ImageJ/Fiji, Imaris, Amira, Optiland, GUI development
Biological Applications	Mouse and zebrafish embryonic development, vascular imaging, prenatal alcohol exposure, developmental toxicity, live-sample imaging, optical clearing

Research Experience

Graduate Research Assistant - Multimodal Optical Imaging <i>Biomedical Optics Laboratory, University of Houston</i>	Jan 2021–May 2026 <i>Houston, TX</i>
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- Designed and integrated a co-registered imaging platform combining a **1051-nm swept-source OCT system** with **one-photon and two-photon LSFM** for simultaneous label-free structural and fluorescence imaging of mouse and zebrafish embryos.
- Achieved approximately **7–7.6 μm OCT axial resolution**, **15 μm lateral resolution**, approximately **100 dB sensitivity**, and approximately **6.9 dB sensitivity roll-off over 4.13 mm**; validated resolution, sensitivity, field of view, contrast, and repeatability.
- Engineered beam delivery, polarization management, galvo scanning, scan-lens interfaces, acquisition timing, camera/laser synchronization, sample positioning, and environmental control for stable multimodal 3D and longitudinal imaging.
- Demonstrated one-photon fluorescence imaging to approximately **250 μm** and two-photon visualization beyond **800 μm** in scattering embryonic tissue while OCT provided whole-embryo morphology.
- Developed depth-resolved OCT attenuation algorithms incorporating noise subtraction, confocal correction, and sensitivity-roll-off compensation; validated performance using multilayer TiO_2 phantoms and applied the method to murine brain development.
- Built MATLAB and Python workflows for OCT reconstruction, multimodal co-registration, morphometry, segmentation, attenuation mapping, 3D visualization, quality control, and standardized image processing.
- Developed a low-cost imaging incubator for long-term OCT-LSFM experiments and led developmental phenotyping studies involving prenatal ethanol exposure and zebrafish vasculature.
- Mentored three Ph.D. students and undergraduate researchers in optical alignment, system troubleshooting, image acquisition, quantitative analysis, experimental design, and scientific presentation.

Graduate Research Assistant - Optical Design and Two-Photon Microscopy <i>3D Biomedical Imaging and Technology Laboratory, GIST</i>	Aug 2018–Dec 2020 <i>Gwangju, South Korea</i>
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- Designed and optimized broadband scan and tube lenses for laser-scanning microscopy using **Zemax OpticStudio**; evaluated wavefront error, Strehl ratio, aberrations, telecentricity, field of view, and F-theta distortion across 680–1600 nm.
- Developed four optimized scan lenses and one tube lens, reducing F-theta distortion from 13.00 to 1.94 and telecentric distortion from 3.52 to 0.4367.
- Supported development and qualification of a two-photon laser-scanning microscope through optical component evaluation, alignment, testing, measured-performance analysis, and structured troubleshooting.

Education

Ph.D., Biomedical Engineering

University of Houston

Jan 2021–May 2026

Houston, TX

Advisor: Dr. Kirill V. Larin. Dissertation: *OCT-LS: A multimodal co-registered optical coherence tomography and light-sheet fluorescence microscopy platform for simultaneous structural and molecular imaging of embryonic development.*

M.S., Biomedical Science and Engineering

Gwangju Institute of Science and Technology

Aug 2018–Dec 2020

Gwangju, South Korea

Advisor: Dr. Hyuk-Sang Kwon. Thesis: *Analysis of an achromatic doublet as a scan lens for laser-scanning microscopy.*

B.S., Electrical and Electronic Engineering

International Islamic University Chittagong

Aug 2013–Aug 2017

Chittagong, Bangladesh

Thesis: *Feasibility analysis and proposal for a 1.3-MW hybrid renewable power plant for Saint Martin's Island, Bangladesh.*

Peer-Reviewed Journal Articles

- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Yogeshwari S. Ambekar, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. “Multimodal optical coherence tomography and two-photon light-sheet fluorescence microscopy for embryo imaging.” *Journal of Biomedical Optics* 30(6), 060501 (2025). DOI: 10.1117/1.JBO.30.6.060501.
- **Md Mobarak Karim**, Achuth Nair, Manmohan Singh, Maryam Hatami, Salavat R. Aglyamov, and Kirill V. Larin. “Depth-resolved attenuation coefficient quantification during murine embryonic brain development.” *Journal of Biophotonics* 18(10), e202500212 (2025). DOI: 10.1002/jbio.202500212.
- Leah A. Lewis, **Md Mobarak Karim**, et al. “Multimodal optical imaging for the assessment of the teratogenic effects of ethanol on zebrafish development.” *Journal of Biomedical Optics* 31(6), 066004 (2026). DOI: 10.1117/1.JBO.31.6.066004.
- Mumtaza Mumu, Ayan Das, Talha Bin Emran, Saikat Mitra, Fahadul Islam, Arpita Roy, **Md Mobarak Karim**, Rajib Das, Moon Nyeo Park, Deepak Chandran, Rohit Sharma, Mayeen Uddin Khandaker, Abubakr M. Idris, and Bonglee Kim. “Fucoxanthin: A promising phytochemical on diverse pharmacological targets.” *Frontiers in Pharmacology* 13, 929442 (2022). DOI: 10.3389/fphar.2022.929442.

Manuscripts in Preparation or Under Review

- **Md Mobarak Karim**, et al. “Long-term imaging of zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy.” Manuscript in preparation.
- James L. Klein, **Md Mobarak Karim**, Christian Zevallos-Delgado, Manmohan Singh, Salavat R. Aglyamov, and Kirill V. Larin. “Tartrazine-based optical clearing for monitoring embryogenesis with optical coherence tomography.” Submitted to *Optics Letters*.

Selected Conference Proceedings

- **Md Mobarak Karim**, Leah A. Lewis, Geethangili Madamanchi, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, Arne C. Lekven, and Kirill V. Larin. “OCT-LS: A co-registered OCT and LSFM platform for simultaneous structural and molecular imaging of embryonic development.” *Proceedings of SPIE* 13841, Multimodal Biomedical Imaging XXI (2026). DOI: 10.1117/12.3080842.
- **Md Mobarak Karim**, Leah A. Lewis, Christian Zevallos-Delgado, Oscar E. Ruiz, Geethangili Madamanchi, Alexander W. Schill, Salavat R. Aglyamov, Arne C. Lekven, and Kirill V. Larin. “Combined structural and molecular zebrafish embryonic imaging on ethanol exposure effects during vasculature development.” *Proceedings of SPIE* 13327, Multiscale Imaging and Spectroscopy VI (2025). DOI: 10.1117/12.3043312.
- **Md Mobarak Karim**, Ruijiao Sun, Oscar E. Ruiz, Leah A. Lewis, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. “Imaging zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy.” *Proceedings of SPIE* 12834, Multimodal Biomedical Imaging XIX (2024). DOI: 10.1117/12.3001234.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, Yogeshwari S. Ambekar, Alexander W. Schill, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. “Multimodal optical coherence tomography and two-photon selective-plane illumination microscopy for embryonic imaging.” *Proceedings of SPIE* 12371, Multimodal Biomedical Imaging XVIII (2023). DOI: 10.1117/12.2649508.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. “Combined optical coherence tomography and light-sheet fluorescence microscopy for embryonic imaging.” *Proceedings of SPIE* 11952, Multimodal Biomedical Imaging XVII (2022). DOI: 10.1117/12.2610091.

Advanced Research Training

- **Multiscale Biophotonics Training Program**, Beckman Laser Institute, University of California, Irvine, 2024. NIH-supported training in fluorescence lifetime imaging, nonlinear microscopy, spectroscopy, and translational biophotonics.
- **CLIMB Biophotonics Summer School**, Beckman Institute, University of Illinois Urbana-Champaign, 2023. Hands-on OCT/OCM system construction, reconstruction, and quantitative imaging.
- **Siegman International School on Lasers**, Poland, 2022. Advanced training in laser physics, ultrafast optics, nonlinear optics, and photonic systems.

Selected Research Contributions

- **Co-registered structural and molecular embryo imaging**: Established an OCT-LSFM architecture that combines label-free, depth-resolved morphology with one- and two-photon fluorescence information in a common coordinate system.
- **Quantitative developmental tissue optics**: Developed and validated depth-resolved attenuation mapping for regional assessment of murine embryonic brain development.
- **Developmental toxicology**: Applied multimodal OCT and LSFM to quantify morphological and vascular phenotypes following embryonic ethanol exposure in zebrafish.
- **Longitudinal imaging infrastructure**: Built controlled sample environments and acquisition workflows for stable live-embryo imaging, including temperature and environmental regulation.
- **Optical design**: Optimized scan- and tube-lens systems for broadband laser-scanning microscopy and evaluated aberration, telecentricity, distortion, and field performance.

Selected Honors and Awards

2025	Cover image featured in <i>Journal of Biomedical Optics</i>
2024	NIH Travel Grant, UCI Multiscale Biophotonics Training Program
2022	Zuegel Family Scholarship, Siegman International School on Lasers
2022	OPTICA Siegman International School on Lasers Grant
2022	SPIE Photonics West Travel Grant
2020	Outstanding Poster Award, Biomedical Research Meeting, South Korea
2018–2020	Korean Government Scholarship for M.S. study at GIST
2013–2017	Undergraduate Merit Award and Scholarship, Department of EEE, IIUC

Mentoring and Teaching

- Mentored three Ph.D. students and several undergraduate researchers in OCT/LSFM alignment, microscopy operation, image acquisition, MATLAB/Python analysis, experimental design, and scientific presentation.
- Trained junior researchers in system calibration, image-quality evaluation, multimodal co-registration, segmentation, and 3D visualization.
- Supported multidisciplinary collaboration among biomedical engineers, optical engineers, developmental biologists, and imaging scientists.

Professional Service and Leadership

- Reviewer for biomedical optics and photonics journals.
- Senior Judge, Texas Science and Engineering Fair - Biomedical Division (2022).
- Executive Member, University of Houston SPIE Student Chapter (2021–Present).
- Member, OPTICA (formerly the Optical Society of America) (2021–Present).
- Founder and former executive member, Bangladesh Robotics Foundation.
- Assistant General Secretary, IIUC EEE Club (2017); Troop Leader, Bangladesh Red Crescent Youth; Leadership Graduate, BYLC Art of Leadership Program (2016).

References

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Associate Professor, ECE
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UH Faculty Profile